

SHTIP, North Macedonia

WMO: 135910

Lat: 41.750N

Lon: 22.183E

Elev: 327

StdP: 97.46

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.4	-6.6	-13.0	1.3	-8.4	-10.3	1.6	-5.1	15.1	11.0	13.1	3.9	2.2	320	0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.5	35.7	20.6	34.1	20.4	32.5	19.9	21.8	32.5	21.2	31.6	20.5	30.5	3.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.9	24.6	17.8	13.3	24.2	17.2	12.8	24.0	65.3	32.5	62.8	31.6	60.5	30.6	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.5	10.7	9.2	DB	-12.7	38.6	4.0	2.1	-15.6	40.1	-17.9	41.4	-20.2	42.6	-23.1	44.1	(4)
(5)			WB	-12.6	23.1	3.6	0.8	-15.2	23.6	-17.3	24.1	-19.3	24.6	-21.9	25.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	1.6	4.8	8.8	13.3	17.7	21.8	24.7	25.1	19.9	14.2	8.8	3.2	(6)	
	DBStd	8.80	4.86	4.22	3.76	3.51	3.16	3.22	3.01	2.76	3.36	3.95	4.11	4.70	(7)	
	HDD10.0	788	262	149	68	11	0	0	0	0	0	12	70	217	(8)	
	HDD18.3	2318	519	378	297	154	49	7	1	0	22	134	287	470	(9)	
	CDD10.0	2139	2	4	29	109	240	355	455	467	298	142	34	5	(10)	
	CDD18.3	628	0	0	0	3	31	112	197	208	70	6	0	0	(11)	
	CDH23.3	6803	0	0	1	37	295	1177	2291	2293	637	72	1	0	(12)	
	CDH26.7	3033	0	0	0	3	56	480	1128	1139	218	8	0	0	(13)	
(14) Wind	WSAvg	3.0	2.8	3.5	3.7	3.6	2.8	2.7	2.9	2.6	2.8	2.6	3.0	2.9	(14)	
(15) Precipitation	PrecAvg	534	36	31	39	52	53	47	35	35	45	57	47	55	(15)	
	PrecMax	913	111	130	105	156	114	114	108	84	182	154	99	132	(16)	
	PrecMin	317	1	3	3	6	25	13	1	3	5	1	11	0	(17)	
	PrecStd	127	23	25	24	34	23	24	25	22	39	39	22	33	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.7	18.3	22.5	26.9	30.6	35.8	38.0	37.9	33.5	27.6	21.2	17.2	(19)	
		MCWB	10.5	11.2	12.3	16.3	18.2	21.0	21.0	21.0	19.8	17.5	14.6	13.5	(20)	
	2%	DB	12.4	15.6	20.0	24.0	28.2	33.5	35.7	36.0	31.4	25.4	18.7	13.9	(21)	
		MCWB	9.2	9.8	11.4	14.9	17.9	20.8	20.6	20.6	18.9	16.2	13.9	11.0	(22)	
	5%	DB	10.4	13.7	17.8	22.0	26.6	31.6	34.1	34.1	29.2	23.2	17.0	11.9	(23)	
		MCWB	7.6	9.1	10.4	13.7	17.2	20.1	20.3	20.3	18.2	15.4	13.4	9.5	(24)	
	10%	DB	8.6	11.8	15.7	20.1	24.8	29.7	32.4	32.4	27.1	21.1	15.4	10.1	(25)	
		MCWB	6.5	8.1	9.4	12.7	16.6	19.3	19.9	19.7	17.4	14.9	12.1	8.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	12.1	13.5	17.1	19.7	22.6	22.6	22.7	21.3	18.4	15.6	13.5	(27)	
		MCDB	14.2	17.1	20.5	25.0	28.4	33.1	34.2	34.2	30.7	25.5	19.4	16.7	(28)	
	2%	WB	9.4	10.5	12.0	15.6	18.6	21.5	21.7	21.7	19.7	17.1	14.4	11.2	(29)	
		MCDB	11.9	14.6	18.0	22.6	26.3	31.4	32.6	32.6	28.5	22.6	17.7	13.3	(30)	
	5%	WB	7.9	9.4	11.1	14.4	17.8	20.6	21.1	21.1	18.8	16.2	13.5	9.9	(31)	
		MCDB	10.1	13.2	16.5	20.7	25.1	29.9	31.7	31.7	26.7	21.3	16.6	11.7	(32)	
	10%	WB	6.6	8.3	10.1	13.3	17.0	19.8	20.4	20.4	18.1	15.4	12.5	8.5	(33)	
		MCDB	8.4	11.6	15.0	18.8	23.7	28.3	30.5	30.4	25.0	20.0	15.3	10.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	8.9	10.2	10.6	11.4	12.4	13.4	13.5	11.9	10.6	8.0	6.4	(35)	
		MCDBR	8.5	12.4	14.6	13.9	14.4	15.3	15.9	15.8	15.3	14.3	10.2	7.6	(36)	
	5% WB	MCWBR	5.8	7.5	7.5	6.5	5.8	5.2	5.0	5.0	5.8	6.7	5.9	5.2	(37)	
		MCWBR	7.3	11.1	12.7	12.6	13.0	14.4	14.4	14.4	12.9	11.5	8.8	6.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.332	0.359	0.404	0.462	0.454	0.438	0.436	0.452	0.431	0.394	0.362	0.335	(40)	
		taud	2.367	2.284	2.186	2.034	2.129	2.221	2.225	2.178	2.211	2.298	2.345	2.360	(41)	
	Ebn at Noon		804	835	835	809	827	841	837	810	798	784	758	764	(42)	
		Edh at Noon	84	106	131	164	153	140	138	140	127	103	85	78	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.67	2.54	3.62	4.65	5.64	6.49	6.73	6.00	4.40	3.07	1.97	1.46	(44)	
	RadStd		0.26	0.33	0.42	0.53	0.37	0.48	0.39	0.39	0.35	0.35	0.20	0.24	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	+0.50	N/A	N/A	N/A	N/A	N/A	N/A	-151	N/A	N/A

Nomenclature: See separate page